UNIVERSITY OF CENTRAL PUNJAB LAHORE

Assignment # 1 Version 1

CLO Mapping: CLO1

Course Code	**MAT243**	**Semester**	**FALL2025**
Course Title	**Multivariate Calculus**		
Resource Person	**SADIA SALMAN**		
Assignment Given Date	**20-10-2025**	**Submission Date**	**25-10-2025**
Total Marks	**25**		

Name of student	Complete Registration no.	Section	Assignment marks	Viva marks	Total marks

Submission Instructions *(Please follow strictly)*

- **Take the PRINTOUT OF THIS ASSIGNMENT PDF file and solve the question at the given empty spaces in the assignment PDF file. DO NOT use any other papers. Alternatively, you can only use simple plane white printer pages.**
- **This is the GROUP ASSIGNMENT of TWO (2) members keeping in view that version 1 will solve one group member and version 2 will solve the other group member. Each group will submit one collective assignment.**
- **It is mandatory for both members to know the method and procedure of solving both questions. Any question may be asked from any member in the viva.**
- **Write your complete names and registration numbers on each page of the assignment.**
- **Follow the deadline. Finish your work one day before, so you could submit in time.**
- **LATE SUBMISSION WILL NOT BE ACCEPTED (Deduction of 5% marks each day).**
- **All Plagiarized assignments will be awarded ZERO marks.**

I declare that I have prepared the assignment according to above guidelines, and I shall be responsible for any deduction of marks if the instructions are not followed.

Student 1 signature: _____ Student 2 signature: _____

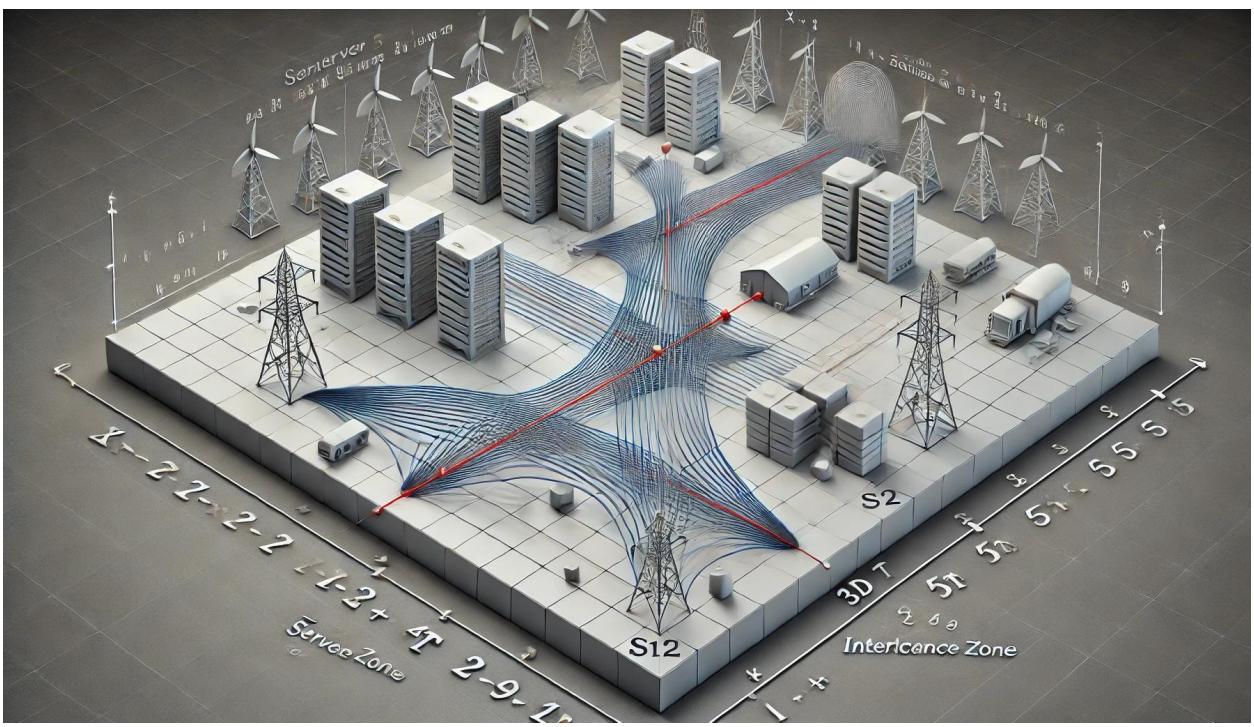
Name: **Registration No.**

Version: 1

Question [CLO1] (25 marks=7+3+7+8): In a **data center**, **network signals** are transmitted between various **servers** positioned in a **3D grid**. A **data transfer path** needs to be **optimized** to avoid **interference zones** created by **signal overlap from other servers**. One of the **main interference zones** is **represented** by a **plane** that **passes through a control point** and **is parallel to the signal route** of another **active network cable**.

The **starting point** of the **primary data transmission route** is at **server** $S_1 = (1, 3, 2)$, and it needs to reach the **target server** $S_2 = (7, 9, 11)$. The **interference zone created by a neighboring server network path is defined by a plane** that **passes through the control point** $P = (4, 5, 6)$ and **is parallel to an active signal path** with the following parametric equations:

$$\begin{cases} x = 3 + t \\ y = 2 + 4t \\ z = -1 + 5t \end{cases}$$



a) **Determine** the **equation of the plane** representing the **interference zone boundary**. This plane passes through the control point $\boldsymbol{P} = (\mathbf{4}, \mathbf{5}, \mathbf{6})$ and is **parallel** to the **neighbouring network path**.

b) **Find** the **equation of the line** that **represents** the **optimal data transmission route** from **server** $\boldsymbol{S_1} = (\mathbf{1}, \mathbf{3}, \mathbf{2})$ to **server** $\boldsymbol{S_2} = (\mathbf{7}, \mathbf{9}, \mathbf{11})$.

c) To avoid interference, **calculate** the **shortest distance from the starting point** $\boldsymbol{S_1} = (\mathbf{1}, \mathbf{3}, \mathbf{2})$ to the **interference zone plane**. If this distance is less than a specified threshold, the route will need adjustment.

d) As an additional safety measure, **calculate** the **distance from the target server** $\boldsymbol{S_2} = (\mathbf{7}, \mathbf{9}, \mathbf{11})$ to the neighbouring network path to ensure it remains free from interference.

Marks Distribution: part a: 7 marks, part b: 3 marks, part c: 7 marks, part d: 8 marks.

Solution:

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Assignment # 1 Version 2

CLO Mapping: CLO1

Course Code	**MAT243**	**Semester**	**Spring 2025**
Course Title	**Multivariate Calculus**		
Resource Person	**Dr. Muhammad Rayees Ahmad**		
Assignment Given Date	**07-04-2025**	**Submission Date**	**11-04-2025**
Total Marks	**25**		

Name of student	Complete Registration no.	Section	Assignment marks	Viva marks	Total marks

Submission Instructions *(Please follow strictly)*

- **Take the PRINTOUT OF THIS ASSIGNMENT PDF file and solve the question at the given empty spaces in the assignment PDF file. DO NOT use any other papers. Alternatively, you can only use simple plane white printer pages.**
- **This is the GROUP ASSIGNMENT of TWO (2) members keeping in view that version 1 will solve one group member and version 2 will solve the other group member. Each group will submit one collective assignment.**
- **It is mandatory for both members to know the method and procedure of solving both questions. Any question may be asked from any member in the viva.**
- **Write your complete names and registration numbers on each page of the assignment.**
- **Follow the deadline. Finish your work one day before, so you could submit in time.**
- **LATE SUBMISSION WILL NOT BE ACCEPTED (Deduction of 5% marks each day).**
- **All Plagiarized assignments will be awarded ZERO marks.**

I declare that I have prepared the assignment according to above guidelines, and I shall be responsible for any deduction of marks if the instructions are not followed.

Student 1 signature: _____ Student 2 signature: _____

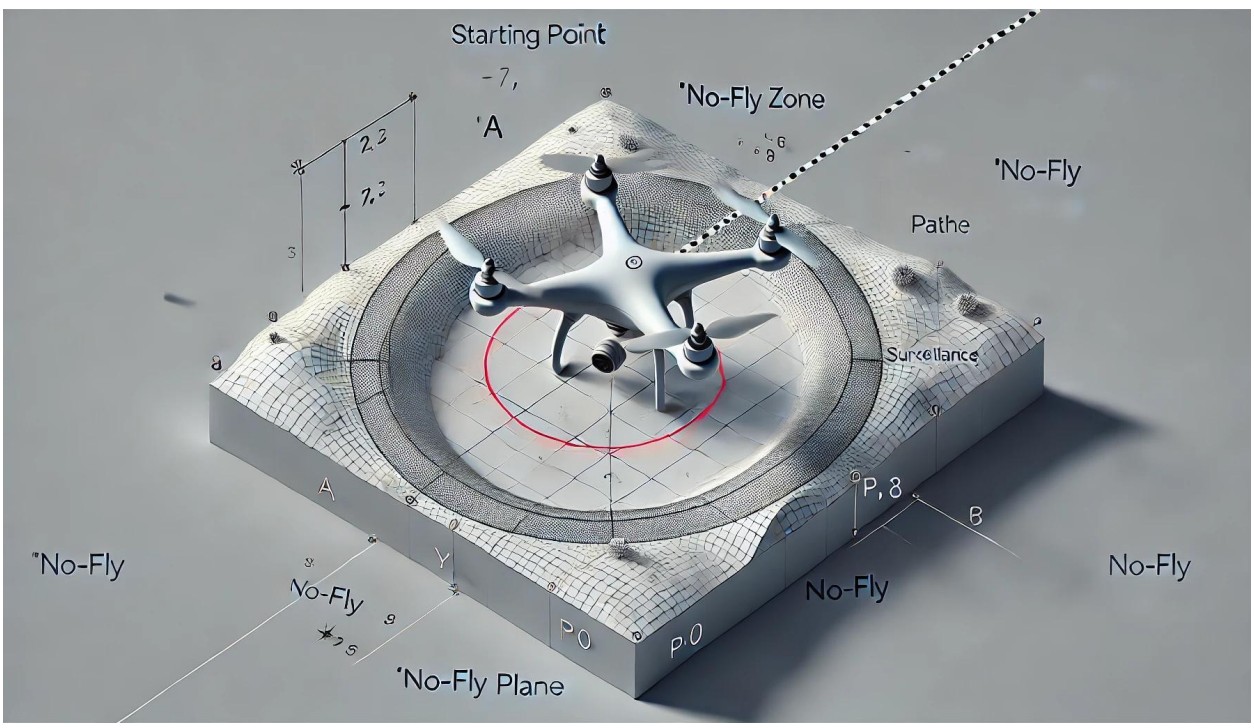
Name: **Registration No.**

Version: 2

Question [CLO1] (25 marks=7+3+7+8): Imagine you are part of a team developing **navigation algorithms for drones** to **deliver packages**. A **drone** must travel from **its starting point to a delivery point** while maintaining a **safe distance** from designated **no-fly zones** defined by **specific planes in space**.

The **no-fly zone** near an **office building** is defined by a **plane** passing through a given point $\boldsymbol{P_0}$ and is **parallel to a line** that **represents** a **surveillance path** with the **parametric equations**:

$$\begin{cases} \boldsymbol{x = 5 + t} \\ \boldsymbol{y = 1 + 3t} \\ \boldsymbol{z = 4t} \end{cases}$$



a) **Determine** the **equation of the no-fly plane** that passes through the **point** $P_0 = (2, 4, 5)$ and is **parallel** to the given **surveillance path**.

b) Assume the **drone's starting point** is at $\boldsymbol{A = (1, 2, 3)}$ and it must **head towards** the **delivery point** $\boldsymbol{B = (6, 7, 8)}$ in a **straight line**. **Find** the **equation of the line** representing the **drone's path**.

c) To ensure the **drone doesn't violate the no-fly zone**, **calculate** the shortest distance between the **drone's starting point** $\boldsymbol{A = (1, 2, 3)}$ and the **no-fly plane**. If this distance is less than a predefined safety distance, **determine** a new **trajectory** by adjusting the **drone's path**.

d) As a secondary safety measure, **compute** the **minimum distance** from the point $\boldsymbol{B = (6, 7, 8)}$ to the **surveillance path**. If this **distance** is below the **threshold**, the **drone** may need to adjust its flight plan to avoid detection.

e) **Marks Distribution: part a: 7 marks, part b: 3 marks, part c: 7 marks, part d: 8 marks.**

Name: **Registration No.**

Solution:

Name: **Registration No.**